

Amdahl's Law

Speedup = 1 / (S + P/N)

Where:

- **P** = Parallelizable fraction of the code
- **S** = Serial (non-parallelizable) fraction of the code ($S = 1 - P$)
- **N** = Number of processors

Occupancy

Example Scenario:

- **Hardware Limits:** 256 max threads, 64 KB register file, 32 KB shared memory per SM.
- **Kernel Requirements:** 32 threads/block, 2 KB shared memory/block, 64 registers/thread.

Let's calculate the block limits based on each constraint:

- **Threads:** 256 max threads / 32 threads per block = **8 blocks**
 - **Registers:** 64 KB (1024 registers of 64 bytes) -> (64 * 1024) / (64 * 32) = 32 blocks.
 - **Shared Memory:** 32 KB / 2 KB per block = **16 blocks**
- The final occupancy is the minimum of all constraints: **8 CUDA blocks per SM**.

SIMT stack frame

SIMT Stack Frame: A hardware structure used to handle SIMT branch divergence inside a warp. Each frame stores:

1. **Re-convergence PC (RPC):** The target instruction address where divergent branch paths merge back.
 2. **Alternate Mask:** A 32-bit mask indicating which thread lanes execute the alternate branch path
 3. **Alternate PC:** The starting address of the alternate branch path code
- Upon hitting a divergent branch, the processor pushes the **alternative PC value** and the **reconvergence point** onto the SIMT stack.
 - When the processor computes the next PC and finds that it matches the reconvergence point at the top of the stack, it **pops** the stack.
 - This process repeats until all alternative paths are executed and the stack is empty for that branch, allowing all threads to finally reconverge.

CPI

Average CPI = CPI_steady_state + sum(Frequency_event * CPI_event)

Example: 5-stage in-order CPU (CPI_steady = 1). Branch mispredict penalty is 3 (2% frequency), Cache miss penalty is 5 (5% frequency).
Average CPI = 1 + (0.02 * 3) + (0.05 * 5)

For multithreading, ideal CPI is scaled by w_depth :
CPI_MT = (CPI_single / w_depth) + Resource_Contentions
Resource contentions include MSHRs, busy ALUs, and DRAM bandwidth limits.

roofline model

- **X-Axis:** Arithmetic Intensity (FLOPs per byte accessed).
- **Y-Axis:** Performance (FLOPs/sec).

Dataflow Analysis

Feature	Reaching Definitions	Live Variables
Domain	Sets of definitions	Sets of variables
Direction	forward	backward
Transfer function $f_b(x)$	$gen_B \cup (x - kill_B)$	$use_B \cup (x - def_B)$
Boundary Condition	$OUT[ENTRY] = \emptyset$	$IN[EXIT] = \emptyset$
Meet Operation ($\wedge$)	$\cup$	$\cup$
Equations	$OUT[B] = f_b(IN[B])$ $IN[B] = \wedge out[pred(b)]$	$IN[B] = f_b(OUT[B])$ $OUT[B] = \wedge in[succ(b)]$
Initialize	$OUT[B] = \emptyset$	$IN[B] = \emptyset$

GPU Support for Multi-Tenant Computing

Mechanisms	Stream	MPS	MIG	
Partition type	No	Logical	Physical	
Max Partition	Unlimited	48	7	
SM isolation	No	By percentage	Yes	
Mem BW isolation	No	No	Yes	
Performance QoS	No	partial	Yes	
Reconfiguration	Dynamic	Process launch	When idle	